



Lab 09: Pub/Sub Qwik Start - Python

Google Cloud Associate Cloud Engineer | GSP094

Overview

Used the Python client library to create Pub/Sub topics and subscriptions, published messages using gcloud commands, and received messages using a Python subscriber script. Combined Python programming with Google Cloud messaging.

Objectives

- Set up a Python virtual environment
- Install the Google Cloud Pub/Sub Python client library
- Create a topic and subscription using Python scripts
- Publish messages to the topic using gcloud
- Pull and view messages using the Python subscriber script

Task 1 - Create a Virtual Environment

```
sudo apt-get install -y virtualenv
python3 -m venv venv
source venv/bin/activate
```

```
(qwiklabs-gcp-01-cda963e43883) x + Editor
Gemini CLI is available in Cloud Shell terminal! Type gemin... Don't show again Dismiss

Selecting previously unselected package python3-distlib.
(Reading database ... 120877 files and directories currently installed.)
Preparing to unpack .../0-python3-distlib_0.3.8-1_all.deb ...
Unpacking python3-distlib (0.3.8-1) ...
Selecting previously unselected package python3-filelock.
Preparing to unpack .../1-python3-filelock_3.13.1-1_all.deb ...
Unpacking python3-filelock (3.13.1-1) ...
Selecting previously unselected package python3-platformdirs.
Preparing to unpack .../2-python3-platformdirs_4.2.0-1_all.deb ...
Unpacking python3-platformdirs (4.2.0-1) ...
Selecting previously unselected package python3-wheel-whl.
Preparing to unpack .../3-python3-wheel-whl_0.42.0-2_all.deb ...
Unpacking python3-wheel-whl (0.42.0-2) ...
Selecting previously unselected package python3-virtualenv.
Preparing to unpack .../4-python3-virtualenv_20.25.0+ds-2_all.deb ...
Unpacking python3-virtualenv (20.25.0+ds-2) ...
Selecting previously unselected package virtualenv.
Preparing to unpack .../5-virtualenv_20.25.0+ds-2_all.deb ...
Unpacking virtualenv (20.25.0+ds-2) ...
Setting up python3-filelock (3.13.1-1) ...
Setting up python3-distlib (0.3.8-1) ...
Setting up python3-platformdirs (4.2.0-1) ...
Setting up python3-wheel-whl (0.42.0-2) ...
Setting up python3-virtualenv (20.25.0+ds-2) ...
Setting up virtualenv (20.25.0+ds-2) ...
Processing triggers for man-db (2.12.0-4build2) ...
student_01_af3616165307@cloudshell:~ (qwiklabs-gcp-01-cda963e43883)$ python3 -m ve
nv venv
student_01_af3616165307@cloudshell:~ (qwiklabs-gcp-01-cda963e43883)$ source venv/b
in/activate
(venv) student_01_af3616165307@cloudshell:~ (qwiklabs-gcp-01-cda963e43883)$
```

Task 2 - Install the Client Library

```
pip install --upgrade google-cloud-pubsub
git clone https://github.com/googleapis/python-pubsub.git
cd python-pubsub/samples/snippets
```

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Editor



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```
----- 84.0/84.0 kB 6.7 MB/s eta 0:00:00
Downloading urllib3-2.7.0-py3-none-any.whl (131 kB)
----- 131.1/131.1 kB 11.5 MB/s eta 0:00:00
Downloading pycparser-3.0-py3-none-any.whl (48 kB)
----- 48.2/48.2 kB 3.5 MB/s eta 0:00:00
Installing collected packages: urllib3, typing-extensions, pycparser, pyasn1, prot
obuf, idna, charset_normalizer, certifi, requests, pyasn1-modules, proto-plus, ope
n telemetry-api, grpcio, googleapis-common-protos, cffi, opentelemetry-semantic-con
ventions, grpcio-status, cryptography, opentelemetry-sdk, grpc-google-iam-v1, goog
le-auth, google-api-core, google-cloud-pubsub
Successfully installed certifi-2026.5.20 cffi-2.0.0 charset_normalizer-3.4.7 crypt
ography-48.0.0 google-api-core-2.30.3 google-auth-2.53.0 google-cloud-pubsub-2.38.
0 googleapis-common-protos-1.75.0 grpc-google-iam-v1-0.14.4 grpcio-1.80.0 grpcio-s
tatus-1.80.0 idna-3.16 opentelemetry-api-1.42.1 opentelemetry-sdk-1.42.1 opentelem
etry-semantic-conventions-0.63b1 proto-plus-1.28.0 protobuf-6.33.6 pyasn1-0.6.3 py
asn1-modules-0.4.2 pycparser-3.0 requests-2.34.2 typing-extensions-4.15.0 urllib3-
2.7.0
(venv) student_01_af3616165307@cloudshell:~ (qwiklabs-gcp-01-cda963e43883)$ git cl
one https://github.com/googleapis/python-pubsub.git
Cloning into 'python-pubsub'...
remote: Enumerating objects: 14460, done.
remote: Counting objects: 100% (272/272), done.
remote: Compressing objects: 100% (157/157), done.
remote: Total 14460 (delta 172), reused 147 (delta 113), pack-reused 14188 (from 3
)
Receiving objects: 100% (14460/14460), 6.79 MiB | 12.68 MiB/s, done.
Resolving deltas: 100% (9678/9678), done.
(venv) student_01_af3616165307@cloudshell:~ (qwiklabs-gcp-01-cda963e43883)$ cd pyt
hon-pubsub/samples/snippets
(venv) student_01_af3616165307@cloudshell:~/python-pubsub/samples/snippets (qwikla
bs-gcp-01-cda963e43883)$
```

Task 3 - Pub/Sub Basics

Explored the publisher.py script to understand available operations.

```
cat publisher.py
python publisher.py -h
```

```
(venv) student_01_af3616165307@cloudshell:~/python-pubsub/samples/snippets (qwikla
bs-gcp-01-cda963e43883)$ python publisher.py -h
usage: publisher.py [-h]
                    project_id
                    {list,create,pubsub-publish-otel-tracing,create_kinesis_ingest
ion,create_cloud_storage_ingestion,create_aws_msk_ingestion,create_azure_event_hub
s_ingestion,create_confluent_cloud_ingestion,create_smt,update_kinesis_ingestion,d
elete,publish,publish-with-custom-attributes,publish-with-error-handler,publish-wi
th-batch-settings,publish-with-flow-control,publish-with-retry-settings,publish-wi
th-ordering-keys,resume-publish-with-ordering-keys,detach-subscription}
                    ...

This application demonstrates how to perform basic operations on topics
with the Cloud Pub/Sub API.

For more information, see the README.md under /pubsub and the documentation
at https://cloud.google.com/pubsub/docs.

positional arguments:
  project_id            Your Google Cloud project ID
  {list,create,pubsub-publish-otel-tracing,create_kinesis_ingestion,create_cloud_s
storage_ingestion,create_aws_msk_ingestion,create_azure_event_hubs_ingestion,create
_confluent_cloud_ingestion,create_smt,update_kinesis_ingestion,delete,publish,publ
ish-with-custom-attributes,publish-with-error-handler,publish-with-batch-settings,
publish-with-flow-control,publish-with-retry-settings,publish-with-ordering-keys,r
esume-publish-with-ordering-keys,detach-subscription}
  list                  Lists all Pub/Sub topics in the given project.
  create                Create a new Pub/Sub topic.
  pubsub-publish-otel-tracing
                        Publish to `topic_id` in `topic_project_id` with
                        OpenTelemetry enabled. Export the OpenTelemetry traces
```

Task 4 - Create a Topic

```
echo $GOOGLE_CLOUD_PROJECT
python publisher.py $GOOGLE_CLOUD_PROJECT create MyTopic
python publisher.py $GOOGLE_CLOUD_PROJECT list
```

```
publish-with-retry-settings    Publishes messages with custom retry settings.
publish-with-ordering-keys     Publishes messages with ordering keys.
resume-publish-with-ordering-keys
                                Resume publishing messages with ordering keys when
                                unrecoverable errors occur.
detach-subscription            Detaches a subscription from a topic and drops all
                                messages retained in it.

options:
  -h, --help                    show this help message and exit
(venv) student_01_af3616165307@cloudshell:~/python-pubsub/samples/snippets (qwikla
bs-gcp-01-cda963e43883)$ python publisher.py $GOOGLE_CLOUD_PROJECT create MyTopic
Created topic: projects/qwiklabs-gcp-01-cda963e43883/topics/MyTopic
(venv) student_01_af3616165307@cloudshell:~/python-pubsub/samples/snippets (qwikla
bs-gcp-01-cda963e43883)$ python publisher.py $GOOGLE_CLOUD_PROJECT list
name: "projects/qwiklabs-gcp-01-cda963e43883/topics/MyTopic"
message_storage_policy {
  allowed_persistence_regions: "us-east4"
}
(venv) student_01_af3616165307@cloudshell:~/python-pubsub/samples/snippets (qwikla
bs-gcp-01-cda963e43883)$
```

Task 5 - Create a Subscription

```
python subscriber.py $GOOGLE_CLOUD_PROJECT create MyTopic MySub
python subscriber.py $GOOGLE_CLOUD_PROJECT list-in-project
python subscriber.py -h
```

```
ySub"
topic: "projects/qwiklabs-gcp-01-cda963e43883/topics/MyTopic"
push_config {
}
ack_deadline_seconds: 10
message_retention_duration {
  seconds: 604800
}
expiration_policy {
  ttl {
    seconds: 2678400
  }
}
state: ACTIVE

(venv) student_01_af3616165307@cloudshell:~/python-pubsub/samples/snippets (qwiklabs-gcp-01-cda963e43883)$ python subscriber.py $GOOGLE_CLOUD_PROJECT list-in-project
projects/qwiklabs-gcp-01-cda963e43883/subscriptions/MySub
(venv) student_01_af3616165307@cloudshell:~/python-pubsub/samples/snippets (qwiklabs-gcp-01-cda963e43883)$ python subscriber.py -h
usage: subscriber.py [-h]
                    project_id
                    {list-in-topic,list-in-project,otel-subscribe,create,create-with-dead-letter-policy,create-push,create-push-no-wrapper,create-with-ordering,create-with-filtering,create-with-exactly-once,create-biquery,create-cloudstorage,create-with-smt,delete,update-push,update-dead-letter-policy,remove-dead-letter-policy,optimistic-subscribe,receive,receive-custom-attributes,receive-flow-control,receive-blocking-shutdown,receive-messages-with-exactly-once-delivery-enabled,receive-synchronously,receive-synchronously-with-lease,listen-for-errors,receive-messages-with-delivery-attempts,receive-messages-with-concurrency-control}
```

Task 6 - Publish Messages

```
gcloud pubsub topics publish MyTopic --message "Hello"
gcloud pubsub topics publish MyTopic --message "Publisher's name is [YOUR NAME]"
gcloud pubsub topics publish MyTopic --message "Publisher likes to eat [FOOD]"
gcloud pubsub topics publish MyTopic --message "Publisher thinks Pub/Sub is awesome"
```

Task 7 - View Messages

```
python subscriber.py $GOOGLE_CLOUD_PROJECT receive MySub
```

Press Ctrl+C to stop listening after messages are received.

```
(venv) student_01_af3616165307@cloudshell:~/python-pubsub/samples/snippets (qwiklabs-gcp-01-cda963e43883)$ python subscriber.py $GOOGLE_CLOUD_PROJECT receive MySub
Listening for messages on projects/qwiklabs-gcp-01-cda963e43883/subscriptions/MySub
b..

Received Message {
  data: b"Publisher's name is Keniyah"
  ordering_key: ''
  attributes: {}
}.
Received Message {
  data: b'Publisher likes to eat apples'
  ordering_key: ''
  attributes: {}
}.
Received Message {
  data: b"Publisher's name is <YOUR NAME>"
  ordering_key: ''
  attributes: {}
}.
Received Message {
  data: b'Publisher thinks Pub/Sub is awesome'
  ordering_key: ''
  attributes: {}
}.
Received Message {
  data: b'Hello'
  ordering_key: ''
  attributes: {}
}.
□
```

Real World Application

- Python microservices use the Pub/Sub client library to subscribe to event streams
- Data pipelines written in Python consume Pub/Sub messages for real-time processing
- ML models trigger automatically when new training data is published to a topic
- Python scripts process IoT sensor data streamed through Pub/Sub

What I Learned

- The Google Cloud Python client library simplifies Pub/Sub operations
- publisher.py and subscriber.py scripts handle topic and subscription management
- Messages can be published via gcloud and received via Python in the same workflow
- The receive command listens continuously until stopped with Ctrl+C
- GOOGLE_CLOUD_PROJECT environment variable stores the project ID automatically

Resources

GCP Pub/Sub Python Client Docs: <https://cloud.google.com/pubsub/docs/publish-receive-messages-client-library>
Lab ID: GSP094